

Insert new clauses Clause 169 to Clause 173 as follows:

169. Introduction to 800 Gb/s networks

169.1 Overview

169.1.1 Scope

This clause describes the general requirements for 800 Gigabit Ethernet.

800 Gigabit Ethernet uses the IEEE 802.3 MAC sublayer operating at a data rate of 800 Gb/s, coupled with any IEEE 802.3 800GBASE Physical Layer implementation and is defined for full duplex operation only.

169.1.2 Relationship of 800 Gigabit Ethernet to the ISO OSI reference model

800 Gigabit Ethernet couples the IEEE 802.3 MAC to a family of 800 Gb/s Physical Layer devices. The relationships among 800 Gigabit Ethernet, the IEEE 802.3 MAC, and the ISO Open System Interconnection (OSI) reference model are shown in Figure 169–1.

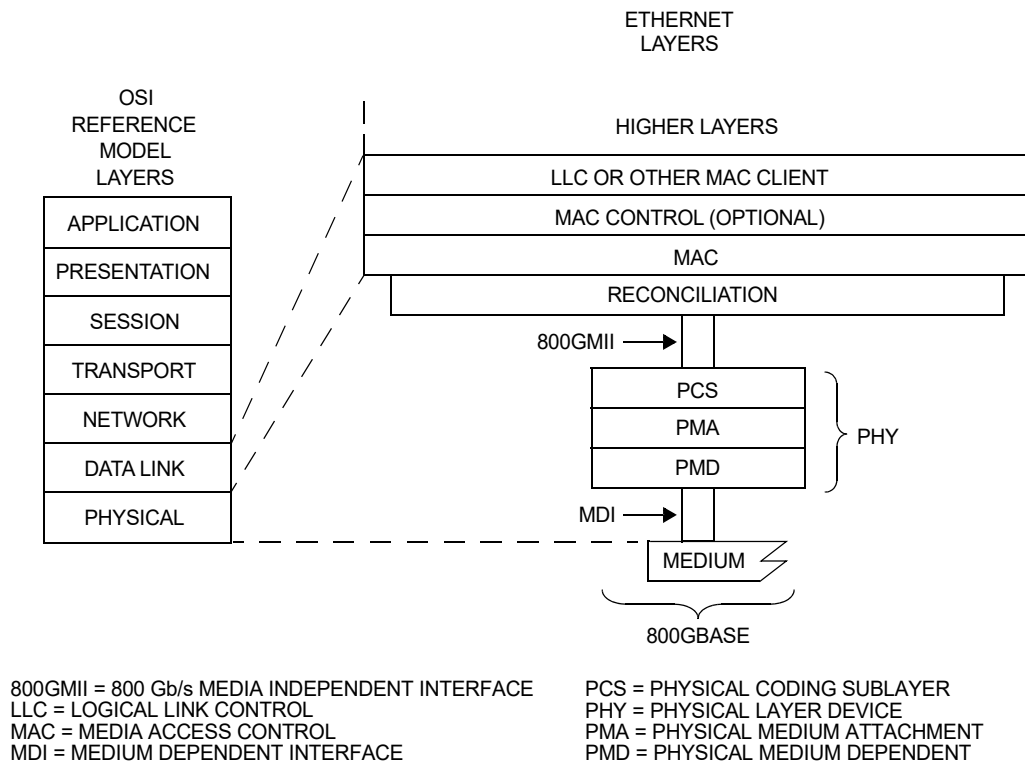


Figure 169–1—Architectural positioning of 800 Gigabit Ethernet

While this specification defines interfaces in terms of bits, octets, and frames, implementations may choose other data-path widths for implementation convenience. The only exceptions are as follows:

- The 800GMII as specified in Clause 170, when implemented as a logical interconnection point between the MAC sublayer and the Physical Layer device (PHY), uses a 64-bit wide data path. Physical instantiations of this interface may use other data-path widths.

- b) The management interface, when physically implemented as the MDIO/MDC (Management Data Input/Output and Management Data Clock) at an observable interconnection port, as specified in Clause 45, uses a bit-wide data path.
- c) The PMA service interface, when physically implemented as 800GAUI-8 (800 Gb/s eight-lane Attachment Unit Interface) at an observable interconnection port, as specified in Annex 120F or Annex 120G, uses an 8-lane data path.
- d) The MDI for each of the following PMDs uses an 8-lane data path:
 - 800GBASE-KR8 as specified in Clause 163
 - 800GBASE-CR8 as specified Clause 162
 - 800GBASE-VR8 and 800GBASE-SR8 as specified in Clause 167
 - 800GBASE-DR8 as and 800GBASE-DR8-2 as specified in Clause 124

169.1.3 Nomenclature

The nomenclature employed by the 800 Gb/s Physical Layer is explained as follows.

The alpha-numeric prefix 800GBASE in the port type (e.g., 800GBASE-R) represents a family of Physical Layer devices operating at a speed of 800 Gb/s.

The term 800GBASE-R represents a family of Physical Layer devices using the Physical Coding Sublayer (PCS) defined in Clause 172 for 800 Gb/s operation.

Physical Layer devices listed in Table 169–1 are defined for operation at 800 Gb/s.

Table 169–1—800 Gb/s PHYs

Name	Description
800GBASE-KR8	800 Gb/s PHY using 800GBASE-R encoding over eight lanes of an electrical backplane (see Clause 163)
800GBASE-CR8	800 Gb/s PHY using 800GBASE-R encoding over eight lanes of twinaxial copper cable (see 1.4.559 and Clause 162)
800GBASE-VR8	800 Gb/s PHY using 800GBASE-R encoding over eight multimode fibers in each direction with reach up to at least 50 m (see Clause 167)
800GBASE-SR8	800 Gb/s PHY using 800GBASE-R encoding over eight multimode fibers in each direction with reach up to at least 100 m (see Clause 167)
800GBASE-DR8	800 Gb/s PHY using 800GBASE-R encoding over eight single-mode fibers in each direction with reach up to at least 500 m (see Clause 124)
800GBASE-DR8-2	800 Gb/s PHY using 800GBASE-R encoding over eight single-mode fibers in each direction with reach up to at least 2 km (see Clause 124)

169.1.4 Physical Layer signaling systems

This standard specifies a family of Physical Layer implementations. Table 169–2 and Table 169–3 specify the correlation between PHY types and clauses. Implementations conforming to one or more PHY types meet the requirements of the corresponding clauses.

Table 169–2—PHY type and clause correlation (800GBASE copper)

PHY type	Clause ^a							
	73	170		171	172	173	120F	162
	Auto-Negotiation	RS	800GMII	800GMII Extender	800GBASE-R PCS	800GBASE-R PMA	800GAUI-8 C2C	800GBASE-CR8 PMD
800GBASE-KR8	M	M	O	O	M	M	O	
800GBASE-CR8	M	M	O	O	M	M	O	M

^a O = Optional, M = Mandatory.

Table 169–3—PHY type and clause correlation (800GBASE optical)

PHY type	Clause ^a									
	170		171	172	173	120F	120G	124		167
	RS	800GMII	800MII Extender	800GBASE-R PCS	800GBASE-R PMA	800GAUI-8 C2C	800GAUI-8 C2M	800GBASE-DR8 PMD	800GBASE-DR8-2 PMD	800GBASE-VR8 PMD
800GBASE-VR8	M	O	O	M	M	O	O			M
800GBASE-SR8	M	O	O	M	M	O	O			M
800GBASE-DR8	M	O	O	M	M	O	O	M		
800GBASE-DR8-2	M	O	O	M	M	O	O		M	

^a O = Optional, M = Mandatory.

169.2 Summary of 800 Gigabit Ethernet architecture

169.2.1 Reconciliation Sublayer (RS) and 800 Gb/s Media Independent Interface (800GMII)

The 800 Gb/s Media Independent Interface (800GMII) specified in Clause 170 provides a logical interconnection between the MAC sublayer and Physical Layer devices (PHYs). The 800GMII is not intended to be physically instantiated, rather it can logically connect layers within a device.

The 800GMII supports 800 Gb/s operation through its 64-bit-wide transmit and receive data paths. The Reconciliation Sublayer (RS) provides a mapping between the signals provided at the 800GMII and the MAC/PLS service definition.

While the 800GMII is an optional interface, it is used extensively in this standard as a basis for functional specification and it provides a common service interface for the Physical Coding Sublayer (Clause 172).

169.2.2 800GMII Extender and 800GMII Extender Sublayer (800GXS)

The optional 800GMII Extender (Clause 171) may be inserted between the Reconciliation Sublayer and the PHY to transparently extend the reach of the 800GMII.

The 800GMII Extender Sublayer (800GXS) is part of the 800GMII Extender. It is equivalent in function to the 800GBASE-R PCS in Clause 172 with the exceptions as defined in Clause 171.

169.2.3 Physical Coding Sublayer (PCS)

The PCS performs encoding of data from the 800GMII, applies FEC, distributes the data to multiple PCS lanes, and transfers the encoded data to the PMA.

The 800GBASE-R PCS is specified in Clause 172.

169.2.4 Physical Medium Attachment (PMA) sublayer

The PMA sublayer provides a medium-independent means to support the use of a range of physical media.

The 800GBASE-R PMA is specified in Clause 173.

169.2.5 Physical Medium Dependent (PMD) sublayer

The PMD sublayer is responsible for interfacing to the transmission medium. The PMD is located just above the Medium Dependent Interface (MDI).

The MDI, logically subsumed within each PMD subclause, is the actual medium attachment for the various supported media.

800GBASE-R PMDs, including the corresponding MDI and media, are specified in Clause 124, Clause 162, Clause 163, and Clause 167.

169.2.6 Auto-Negotiation

Auto-Negotiation provides a device with the capability to detect the abilities (modes of operation) supported by the device at the other end of the link, determine common abilities, and configure for joint operation.

Auto-Negotiation used by the 800 Gb/s backplane PHY (800GBASE-KR8) and the 800 Gb/s copper PHY (800GBASE-CR8) is specified in Clause 73.

169.2.7 Management interface (MDIO/MDC)

The optional MDIO/MDC management interface provides an interconnection between MDIO Manageable Devices (MMDs) and Station Management (STA) entities.

The MDIO/MDC management interface is specified in Clause 45.

169.2.8 Management

Managed objects, attributes, and actions are defined for all 800 Gigabit Ethernet components. These items are specified in Clause 30.

169.3 Service interface specification method and notation

The service interface specification for the 800GBASE-R Physical Layers is as per the definition in 1.2.2. Note that the 800GBASE-R inter-sublayer service interfaces use multiple scalar REQUEST and INDICATION primitives, to indicate the transfer of multiple independent streams of data units, as defined in 169.3.1 through 169.3.3.

169.3.1 Inter-sublayer service interface

The inter-sublayer service interface is described in an abstract manner and does not imply any particular implementation. The inter-sublayer service interface primitives are defined as follows:

IS_UNITDATA_*i*.request
IS_UNITDATA_*i*.indication
IS_SIGNAL.indication
IS_SIGNAL.request

The IS_UNITDATA_*i*.request (where $i = 0$ to $n - 1$, and n is the number of streams of data units) primitive is used to define the transfer of multiple streams of data units to a sublayer from the next higher (closer to the MAC) sublayer.

The IS_UNITDATA_*i*.indication (where $i = 0$ to $n - 1$, and n is the number of streams of data units) primitive is used to define the transfer of multiple streams of data units from a sublayer to the next higher sublayer.

The IS_SIGNAL.indication primitive is used to define the transfer of signal status from a sublayer to the next higher sublayer.

The IS_SIGNAL.request primitive is used to define the transfer of signal status to a sublayer from the next higher sublayer.

169.3.2 Instances of the inter-sublayer service interface

The inter-sublayer service interface may be instantiated between different sublayers, hence a prefix notation is defined to identify a specific instance of an inter-sublayer service interface. The following prefixes are defined:

- PMA:—for primitives issued on the interface between the PMA and the PCS or DTE 800GXS above called the PMA service interface.
- PHY_XS:—for primitives issued on the interface between the PHY 800GXS and the PMA above called the PHY 800GXS service interface.

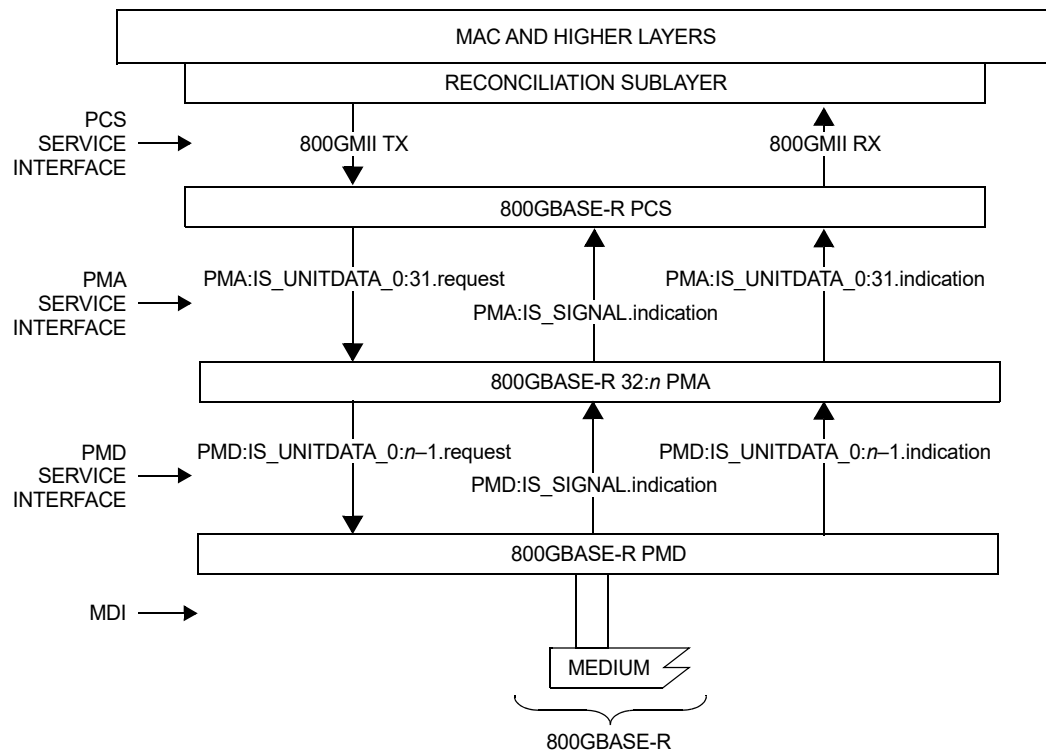
- PMD:—for primitives issued on the interface between the PMD and the PMA above called the PMD service interface.

As an example, the primitives for the PMD service interface are identified as follows:

PMD:IS_UNITDATA_0*i*.request
PMD:IS_UNITDATA_0*i*.indication
PMD:IS_SIGNAL.indication

Examples of inter-sublayer service interfaces for 800GBASE-R with their corresponding instance names are illustrated in Figure 169–2 and Figure 169–3.

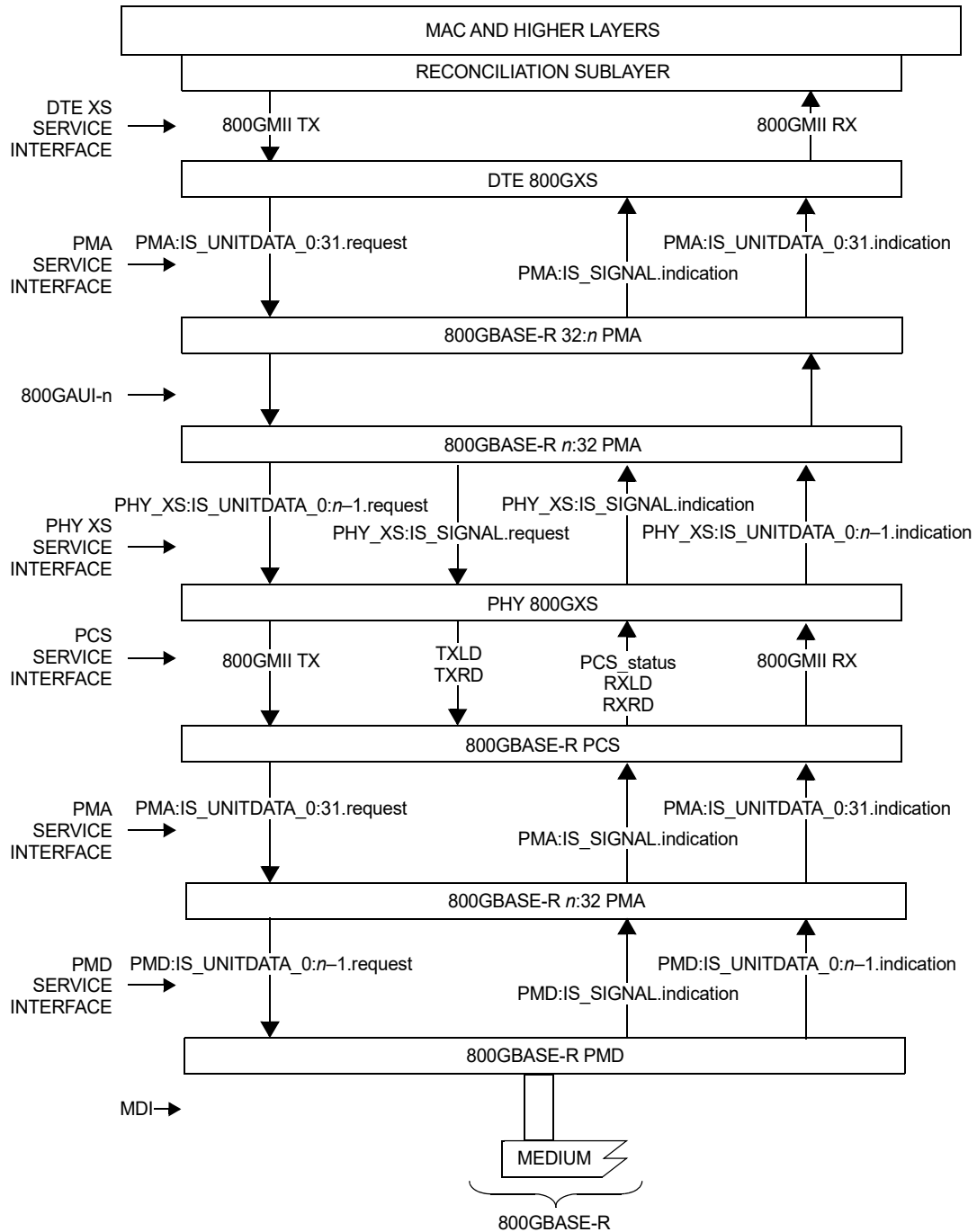
Primitives for other instances of inter-sublayer interfaces are represented in a similar manner.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
n = NUMBER OF PARALLEL STREAMS OF DATA UNITS

Figure 169–2—800GBASE-R inter-sublayer service interfaces not including 800GMII Extender



800GAUI-n = 800 Gb/s ATTACHMENT UNIT INTERFACE
800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
800GXS = 800GMII EXTENDER SUBLAYER
DTE = DATA TERMINAL EQUIPMENT
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
XS = EXTENDER SUBLAYER
n = NUMBER OF PARALLEL STREAMS OF DATA UNITS

Figure 169-3—800GBASE-R inter-sublayer service interfaces including 800GMII Extender

169.3.3 Semantics of inter-sublayer service interface primitives

The semantics of the inter-sublayer service interface primitives for the 800GBASE-R sublayers are described in 116.3.3.1 through 116.3.3.3.

169.4 Delay constraints

Predictable operation of the MAC Control PAUSE operation (Clause 31, Annex 31B) demands that there be an upper bound on the propagation delays through the network. This implies that MAC, MAC Control sublayer, and PHY implementations conform to certain delay maxima, and that network planners and administrators conform to constraints regarding the cable topology and concatenation of devices. Table 169–4 contains the values of maximum delay (sum of transmit and receive delays at one end of the link) for each instance of a sublayer. If a PHY contains an Auto-Negotiation sublayer, the delay of the Auto-Negotiation sublayer is included within the delay of the PMD and medium. The delay for a set of sublayers within the same package may be constrained by the sum of constraints for the set of sublayers.

Table 169–4—Sublayer delay constraints

Sublayer	Maximum (bit time) ^a	Maximum (pause_quantum) ^b	Maximum (ns)	Notes ^c
800G MAC, RS, and MAC Control	196 608	384	245.76	See 170.1.4.
800GBASE-R PCS or 800GXS ^d	640 000	1250	800	See 172.5.
800GBASE-R 32:8 PMA or 8:32 PMA	36 864	72	46.08	See 173.5.4.
800GBASE-R 8:8 PMA	73 728	144	92.16	See 173.5.4.
800GBASE-KR8 PMD	32 768	64	40.96	Includes allocation of 14 ns for one direction through backplane medium. See 163.5.
800GBASE-CR8 PMD	32 768	64	40.96	Includes allocation of 14 ns for one direction through cable medium. See 162.5.
800GBASE-VR8 PMD	16 384	32	20.48	Includes 2 m of fiber. See 167.3.1.
800GBASE-SR8 PMD	16 384	32	20.48	Includes 2 m of fiber. See 167.3.1.
800GBASE-DR8 PMD	16 384	32	20.48	Includes 2 m of fiber. See 124.3.1.
800GBASE-DR8-2 PMD	16 384	32	20.48	Includes 2 m of fiber. See 124.3.1.

^a For 800GBASE, 1 bit time (BT) is equal to 1.25 ps. (See 1.4.215 for the definition of bit time.)

^b For 800GBASE, 1 pause_quantum is equal to 640 ps. (See 1.4.459 for the definition of pause_quantum.)

^c Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

^d If an implementation includes the 800GMII Extender, the delay associated with the 800GMII Extender includes two 800GXS sublayers.

The physical medium interconnecting two optical PHYs introduces additional delay in a link.

See 80.4 for the calculation of bit time per meter of fiber or electrical cable.

See 31B.3.7 for PAUSE reaction timing constraints for stations at operating speeds of 800 Gb/s.

169.5 Skew constraints

Skew (or relative delay) can be introduced between PCS lanes by both active and passive elements of an 800GBASE-R link. Skew is defined as the difference between the times of the earliest PCS lane and latest PCS lane for the one to zero transition of the alignment marker sync bits. The PCS deskew function (see 172.2.5.1) compensates for all lane-to-lane Skew observed at the receiver. The Skew between the PCS lanes is kept within limits as shown in Table 169–5 so that the transmitted information on the PCS lanes can be reassembled by the receive PCS.

Skew Variation may be introduced due to variations in electrical, thermal, or environmental characteristics. Skew Variation is defined as the change in Skew between any PCS lane and any other PCS lane over the entire time that the link is in operation. From the time the link is brought up, Skew Variation is limited so that each PCS lane always traverses the same PCS lane between any pair of adjacent sublayers while the link remains in operation.

The maximum Skew and Skew Variation at physically instantiated interfaces is specified at Skew points SP1, SP2, and SP3 for the transmit direction and SP4, SP5, and SP6 for the receive direction as illustrated in Figure 169–4 for a PHY with one 800GAUI-n and Figure 169–5 for a PHY with two 800GAUI-n.

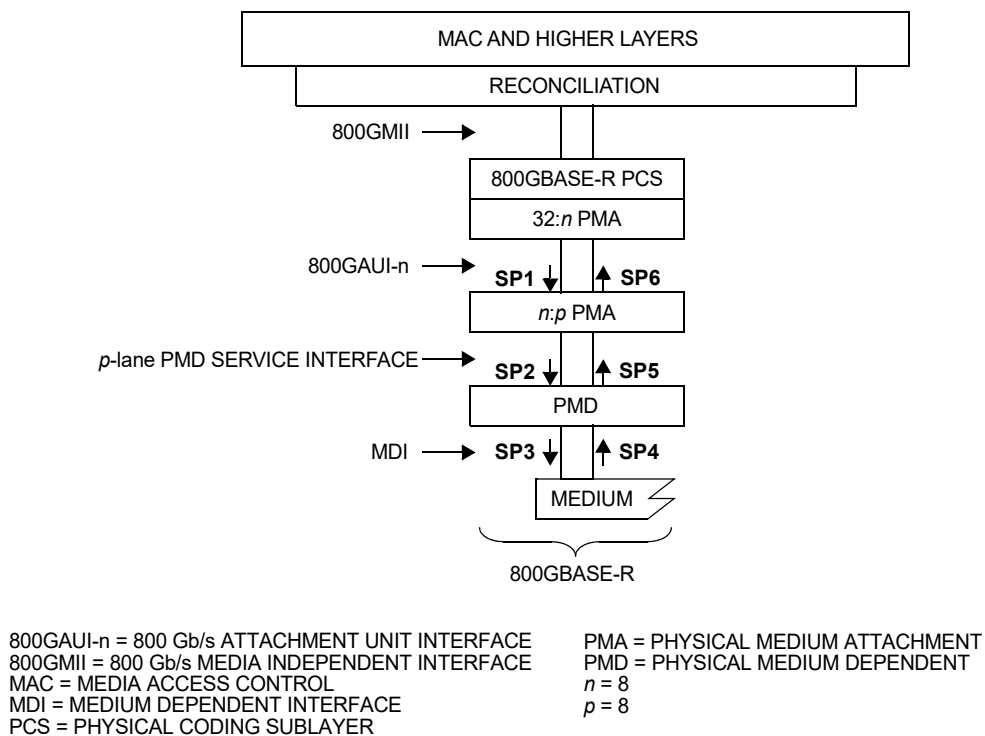
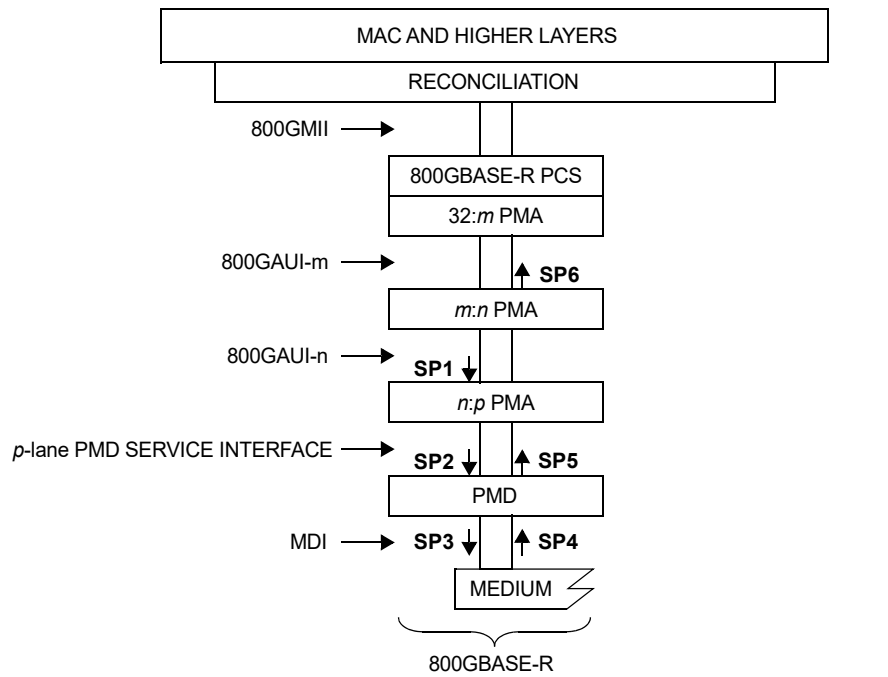


Figure 169–4—800GBASE-R Skew points for a PHY with one 800GAUI-n



800GAUI-n = 800 Gb/s ATTACHMENT UNIT INTERFACE
800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
 $m=8$
 $n=8$
 $p=8$

Figure 169–5—800GBASE-R Skew points for a PHY with two 800GAUI-n

In the transmit direction, the Skew points are defined in the following locations:

- SP1 on the 800GAUI-n interface, at the input of the PMA closest to the PMD
- SP2 on the PMD service interface, at the input of the PMD
- SP3 at the output of the PMD, at the MDI

In the receive direction, the Skew points are defined in the following locations:

- SP4 at the MDI, at the input of the PMD
- SP5 on the PMD service interface, at the output of the PMD
- SP6 on the 800GAUI-n interface, at the output of the PMA closest to the PCS or DTE 800GX

The allowable limits for Skew are shown in Table 169–5 and the allowable limits for Skew Variation are shown in Table 169–6.

The Skew and Skew Variation requirements for the PCS, PMA, and PMD sublayers are specified in the respective clauses as noted in Table 169–5 and Table 169–6.

Table 169–5—Summary of Skew constraints

Skew points	Maximum Skew (ns) ^a	Maximum Skew for 800GBASE-R PCS lane (UI) ^b	Notes ^c
SP1	16	≈ 425	See 173.5.3
SP2	25	≈ 664	See 173.5.3, 124.3.2.2, 162.6.3, 163.6.3, 167.3.2.2
SP3	36	≈ 956	See 173.5.3, 124.3.2.2, 162.6.3, 163.6.3, 167.3.2.2
SP4	116	≈ 3081	See 173.5.3, 124.3.2.2, 162.6.3, 163.6.3, 167.3.2.2
SP5	127	≈ 3373	See 173.5.3, 124.3.2.2, 162.6.3, 163.6.3, 167.3.2.2
SP6	145	≈ 3852	See 173.5.3
At PCS receive	152	≈ 4038	See 172.2.5.1

^a The Skew limit includes 1 ns allowance for PCB traces that are associated with the Skew points.

^b The symbol ≈ indicates approximate equivalent of maximum Skew in UI based on 1 UI equals 37.64706 ps at PCS lane signaling rate of 26.5625 GBd.

^c Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

Table 169–6—Summary of Skew Variation constraints

Skew points	Maximum Skew Variation (ns)	Maximum Skew Variation for 53.125 GBd PMD lane (UI) ^a	Notes ^b
SP1	0.2	≈ 11	See 173.5.3
SP2	0.4	≈ 21	See 173.5.3, 124.3.2.2, 162.6.3, 163.6.3, 167.3.2.2
SP3	0.6	≈ 32	See 173.5.3, 124.3.2.2, 162.6.3, 163.6.3, 167.3.2.2
SP4	3.4	≈ 181	See 173.5.3, 124.3.2.2, 162.6.3, 163.6.3, 167.3.2.2
SP5	3.6	≈ 191	See 173.5.3, 124.3.2.2, 162.6.3, 163.6.3, 167.3.2.2
SP6	3.8	≈ 202	See 173.5.3
At PCS receive	4	≈ 213	See 173.5.3

^a The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 18.82353 ps at PMD lane signaling rate of 53.125 GBd.

^b Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

169.6 FEC Degrade

The FEC degrade feature provides the ability to detect and indicate a degrade condition at the RS-FEC decoder using FEC degrade detection, and to propagate the FEC degrade indication using FEC degrade signaling. The propagation of FEC degrade indications across PCS and XS is described in [116.6](#).

FEC degrade detection and signaling specifications for the 800GXS and 800GBASE-R PCS are summarized in 171.6 and 172.1.4, respectively.

169.7 State diagrams

State diagrams take precedence over text.

The conventions of 1.2 are adopted, along with the extensions listed in 21.5.

Multiple states of a function that have a transition to a common state utilizing different qualifiers (for example, multiple exit conditions to an IDLE or WAIT state) may be indicated by a shared arrow. An exit transition arrow connects to the shared arrow, and the qualifier is met prior to termination of the transition arrow on the shared arrow. The shared arrow has no qualifier.

169.8 Protocol implementation conformance statement (PICS) proforma

The supplier of a protocol implementation that is claimed to conform to any part of IEEE Std 802.3, Clause 170 through Clause 173, and related annexes demonstrates compliance by completing a protocol implementation conformance statement (PICS) proforma.

A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of which capabilities and options of the protocol have been implemented. A PICS is included at the end of each clause as appropriate. Each of the 800 Gigabit Ethernet PICS conforms to the notation and conventions used in 21.6.